

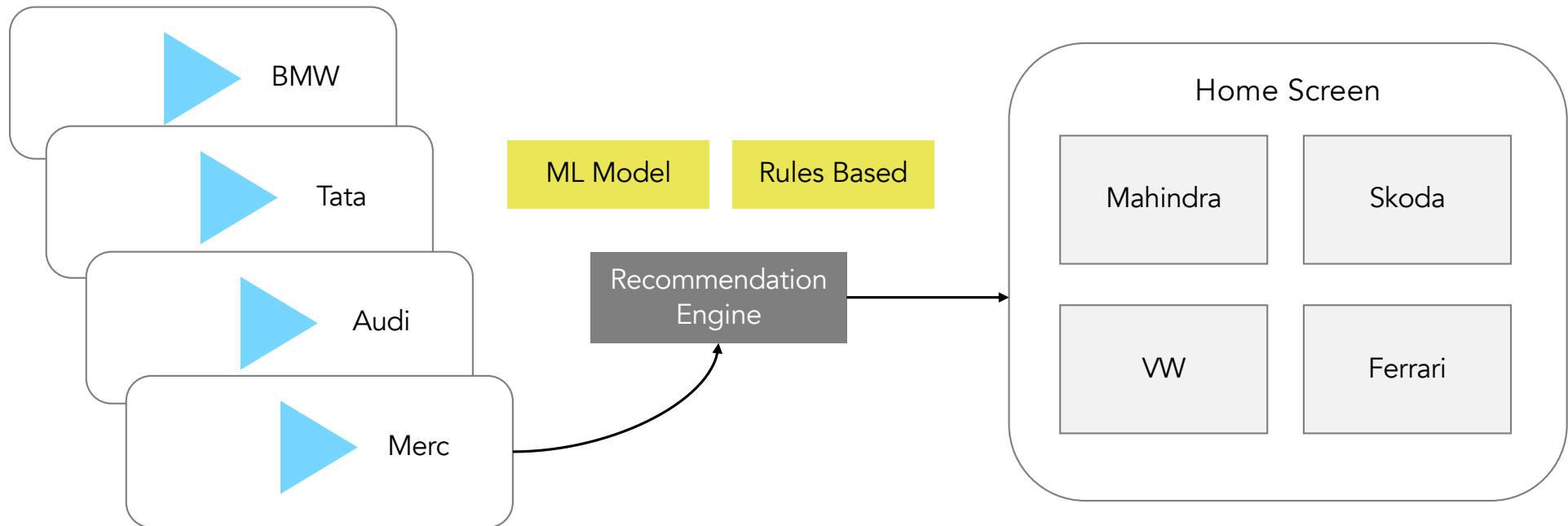
Project 11

Recommendation Engine using Clickstream Data

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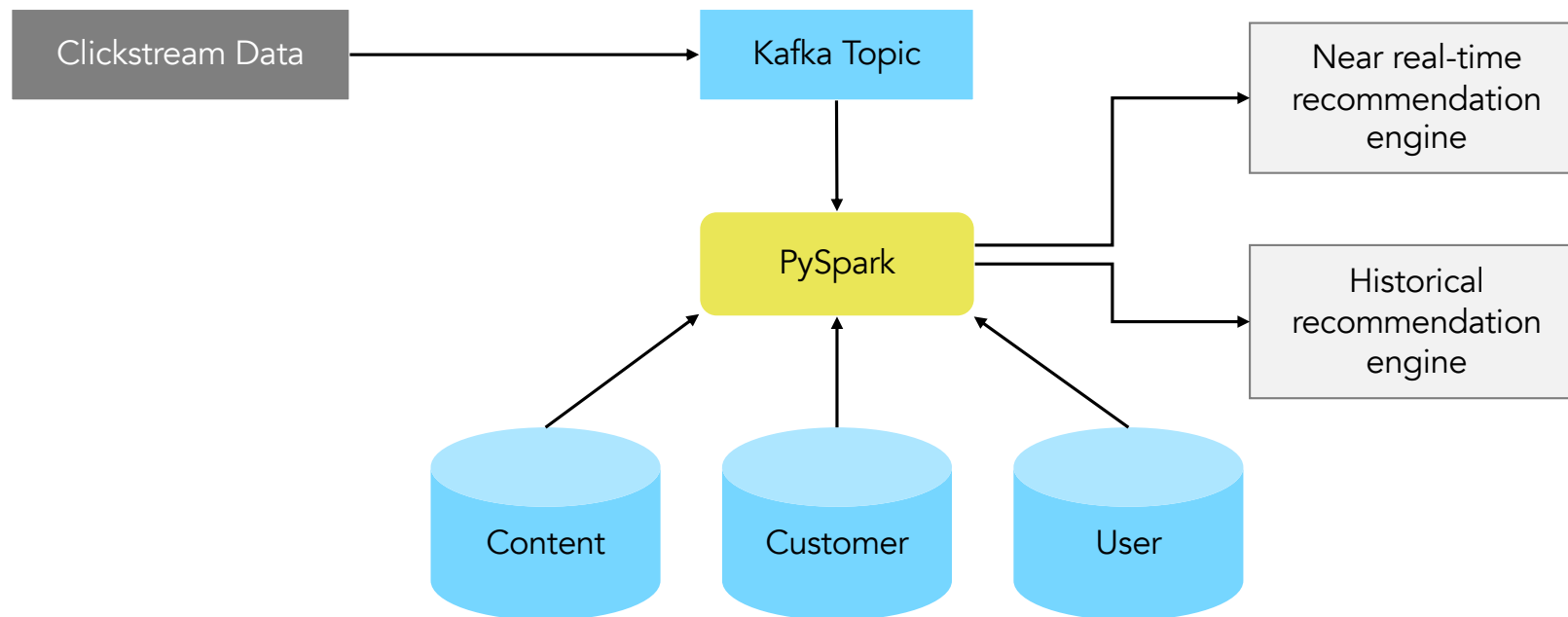
Recommendation Engine

- What happens when you watch same type of videos in YouTube, OTT platforms?

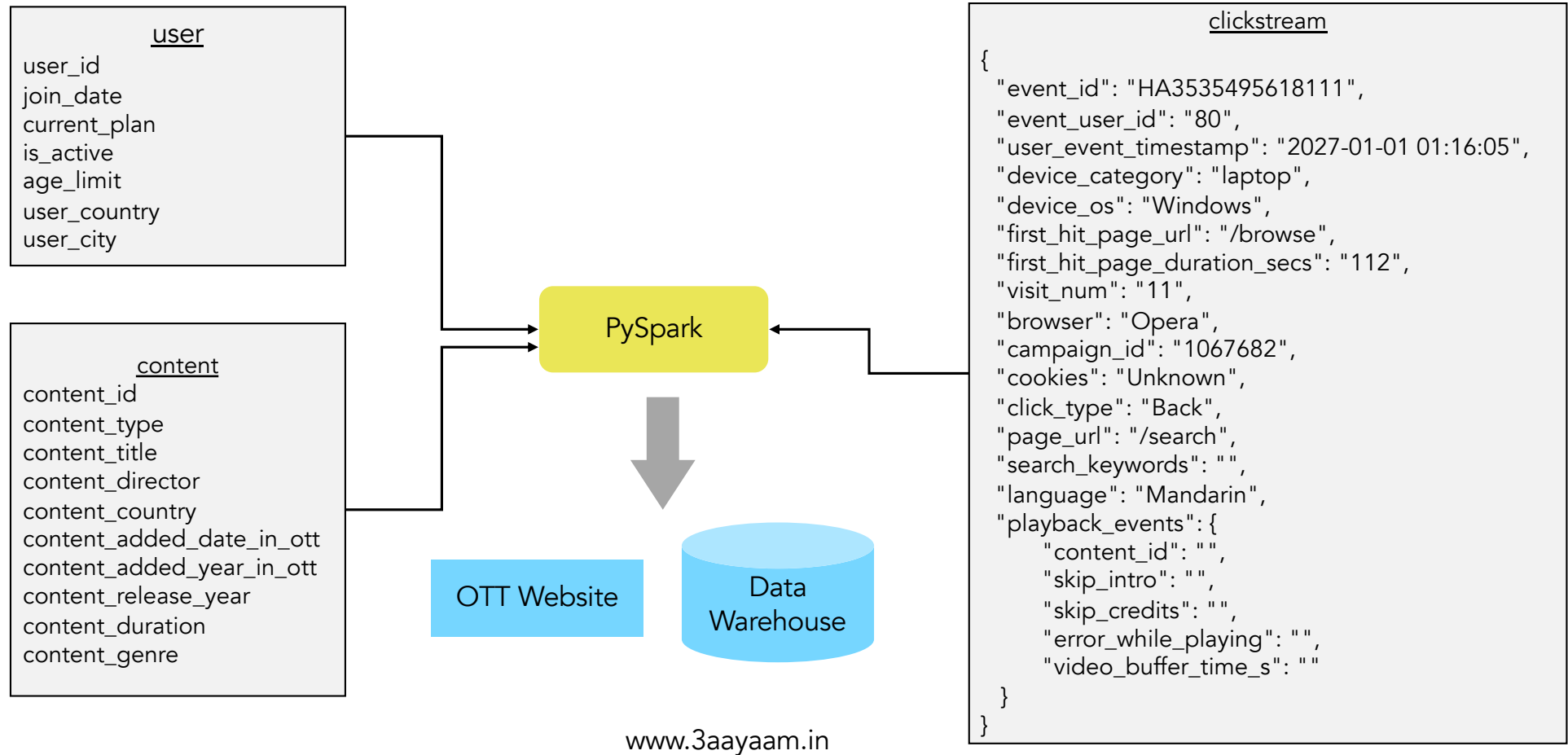


How Recommendation Engine works

- Clicks, URLs, Metadata (Clickstream data) are captured and analyzed.
- Near real-time analytics and historical analytics.



Recommendation Engine



Real-time Recommendation Rules

- Personalized recommendation: Near real-time recommendation should be based on user's browsing, playing video or watching, internal link redirection, searching and downloading contents. These data are captured using clickstream tools.
 - i. Browsing, watching data etc. can be filtered using "click_type" column.
 - ii. Personalized recommendation should be based on last 8 hrs history of every user.
- Generic recommendation: Home page should show top 5 most popular tv shows of that country on a specific day based on previous day's watch/browse/search pattern. This should be shown for all users of that country.
- Generic recommendation: Home page should show top 5 most popular movies of individual genres for all users of a specific country on a specific day based on previous day's watch/browse/search pattern. Should be shown for all users of that country.
- Personalized "because you watched" recommendation: Based on user's "watch" history this should be shown in user's home page.
 - i. Rule : click_type = Play/Pause/Stop and page_url = /watch/xxx

Column Description

Column Name	Details
click_type	AddToCart, Rent, Checkout, Browse , Search , Internal link , External link, Search, Play , Pause , Stop , Back, Download .
page_url (click_type = "Browse")	/genre/drama, /genre/comedy, /genre/fantasy, /genre/action, /genre/cartoon, /genre/sitcom, /genre/crime, /genre/thriller, /genre/adventure, /genre/documentary, /genre/scifi, /genre/history, /genre/romance, /genre/musical
page_url (click_type = "Play", "Pause" or "Stop")	/watch/drama, /watch/comedy, /watch/fantasy, /watch/action, /watch/cartoon, /watch/sitcom, /watch/crime, /watch/thriller, /watch/adventure, /watch/documentary, /watch/scifi, /watch/history, /watch/romance, /watch/musical
page_url (click_type = "Search")	/browse/action, /browse/adventure, /browse/scifi, /browse/thriller, /browse/cartoon, /browse/drama
page_url (click_type = "Download")	/movie/action/, /movie/romance/, /movie/scifi/, /movie/thriller/, /movie/adventure/, /movie/drama/, /tvshow/scifi/, /tvshow/action/, /tvshow/adventure/, /tvshow/anime/, /tvshow/documentary/
page_url (click_type = "Internal Link")	/browse/action, /browse/cartoon, /browse/thriller, /browse/scifi, /browse/adventure

Column Description

Column Name	Details
search_keywords (click_type = "Search")	/search/results?q=scifi, /search/results?q=tom+hanks, /search/results?q=christopher+nolan, /search/results?q=drama+anupam+kher, /search/results?q=action+jason+statham, /search/results?q=action+matt+damon, /search/results?q=angelina+jolie, /search/results?q=romance+matthew+mcconaughey, /search/results?q=comedy+adam+sandler, /search/results?q=action+ajay+devgun, /search/results?q=hindi+top+movies, /search/results?q=koyuki+kato, /search/results?q=brad+pitt, /search/results?q=tom+cruise, /search/results?q=christian+bale

Datasets

- 17 files
- 13 GB (uncompressed)
- ott_cast_data.json, ott_clickstream_data_1-14.json, ott_tv_show_data.csv, ott_user_data.csv, ott_content_data.csv

```
ott_content_data.csv
```

```
content_id,content_type,content_title,content_director,content_country,content_added_date_in_ott,content_added_year_in_ott,content_duration,content_genre
```

```
s2,TV Show,Blood & Water,,South Africa,"September 24, 2021",2021,2 Seasons,"International TV Shows, TV Dramas, TV Mysteries"
```

```
s7,Movie,My Little Pony: A New Generation,"Robert Cullen, José Luis Ucha",,"September 24, 2021",2021,91 min,Children & Family Movies
```


Datasets

```
{"content_id": "s653", "content_cast": ["Nkem Owoh", " Mara Derwent", " Francis Odega", " Cynthia Okereke", " Tom Njemanze", " Romanus Amuta", " Sabina Mole", " Blessing Onolleka", " Oluchi Agunwa", " Chiwendu Onaga", " Patricia Ezeani", " Stephen Ahanonu"]}  
  
{"content_id": "s654", "content_cast": []}  
  
{"content_id": "s655", "content_cast": ["Flavia Hojda", " Crina Semciuc", " Olimpia Melinte", " Sali Levent", " Vlad Logigan", " Alex C\u00f3lin", " Alina Chivulescu", " R\u00f3zvan Vasilescu"]}  
  
{"content_id": "s656", "content_cast": ["Maia Morgenstern", " Olimpia Melinte", " Crina Semciuc", " Flavia Hojda", " Maria Dinulescu", " Alex C\u00f3lin", " Vlad Logigan", " Sali Levent", " Silvia Busuioc", " Mihaela Mihut"]}
```

Historical Recommendation

- The analysis must be done once a month with report generation.
- Content Analysis:
 - Top 5 highest rated TV shows across genres.
 - Which cast and director has the highest content in the channel.
 - Highest searched and browsed content type genre.
 - Which type of movies and TV shows are downloaded the most.
 - Which language has the highest content.
- Customer experience analysis:
 - What was the highest and avg. buffer time of videos for various OS and Browser combination. Does "cookie" configuration have anything to do with it?
 - Which OS and Browser combination had the highest error.
 - What is the distribution of different error types while running across OS and Browser combinations.
 - Which top 3 external OTT platform customers tried to explore.
 - Browser with least error and buffer time.
 - Which type of device is used the most for watching.